AMD Developer Hackathon ACT III

Two Hands

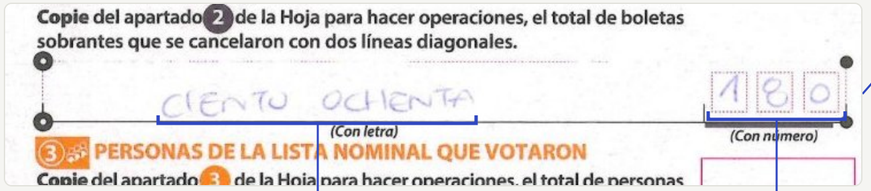
Verifiable quality control for handwritten numbers.

OCR reads the number. Two Hands checks whether the whole form agrees.

For INE, the 32 state institutes and their outside auditors.

UNUSED BALLOTS: TWO RECORDS OF ONE NUMBER

Official typed capture (PREP)	480
Gemma read	180



In words: ciento ochenta (180)
the paper

In digits: 180
the paper



Tally sheet 080310B0100 (INE 2024), privacy-masked.

1,270
real handwritten tally sheets read by Gemma on AMD

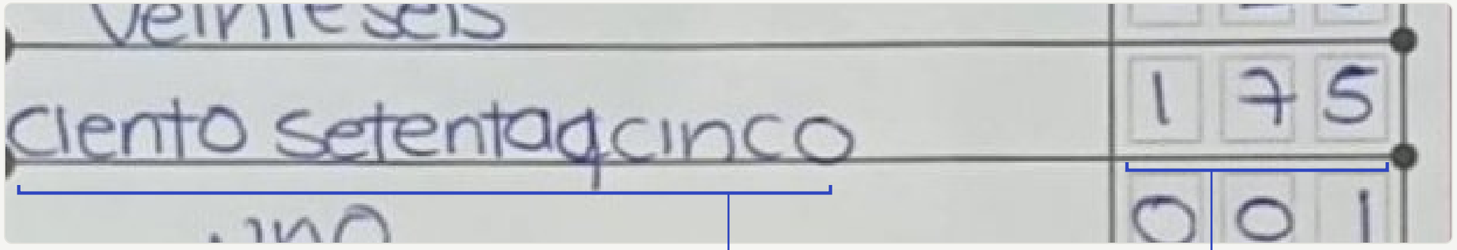
91.6%
field match, vs 62.84% for the reading the state bought

27 of 28
capture errors the certified count reveals, caught (one state, pre-registered)

0 of 484
accepted fields wrong on 100 fresh forms, gate fixed in advance

THE PROBLEM, FROM INE'S OWN FILES

Two official records of the same paper disagree. Neither says which one is right.



In words: ciento setenta y cinco (175)
the paper

In digits: 175
the paper

Form 050815B0100, Option 07 (masked, cropped right of the logo column). One of the 52 on our random 300.

THE TWO TYPED RECORDS OF THIS FIELD

Official typed capture (PREP)	75
Certified count	175

The 75 is a typing error in the database, not on the paper.

6.5%

of the 52,167 polling stations not recounted: the preliminary capture and the certified count disagree on at least one vote field

19.2%

of nationally captured tally forms fail their own row sum (26,336 of 137,196)

Double keying copies the paper faithfully but **cannot say which paper is wrong**. The photo can.

HOW IT WORKS

The model reads. The paper's own arithmetic decides.

01
Photo
Page located and privacy-masked before it is stored or sent

02
Gemma reads
Gemma-4-31B-IT on AMD Token Factory reads every field twice: digits and words

03
Arithmetic
words = digits · rows sum to TOTAL · section 5 = section 7 = TOTAL

04
Accept or route
All hold: **accepted**. One fails: **a person** gets the field, with its photo

05
Receipt
Each field a Merkle leaf; the root anchored with OpenTimestamps

211298B0100

INE 2024 tally sheet, type 2 · phone photo · Puebla · certified by comparison

sample: capture differs from certified

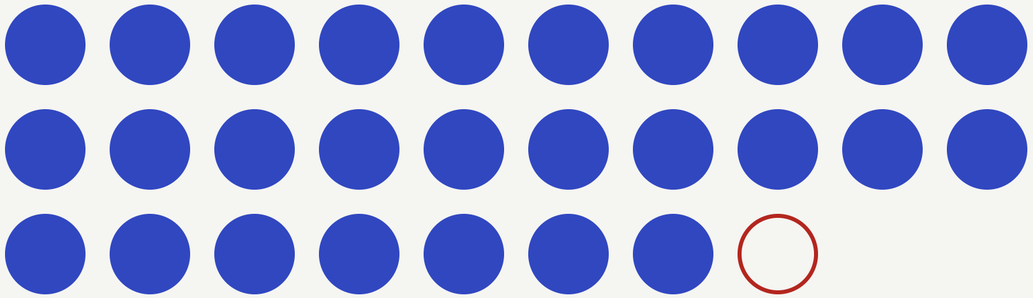
Accepted: 0 fields to key
Every identity on the page holds, so nobody keys this form.

✓	rows sum to TOTAL	417 = 417
✓	votaron + representantes = section 5	417 = 417
✓	section 5 = section 7	417 = 417
✓	section 7 = TOTAL	417 = 417
✓	words = digits on every field	23 of 23

An accepted form on the site (211298B0100): every identity holds, so nobody keys it. A second reader, Qwen3.8-27B on AMD, never overrides Gemma: it can only withhold acceptance.

RESULTS · PRE-REGISTERED · INE'S PUBLIC DATA

1,270 real tally sheets read on AMD. 49 capture errors confirmed by the certified count.



Recall, Colima: each dot is a capture error the certified count reveals. Filled: caught. Outlined: missed. One state; not total recall.

1,270

handwritten tally sheets read by Gemma on AMD

49

capture errors confirmed by the certified count, each with its photo

27 of 28

caught, 96% (Wilson 95% 82-99%), pre-registered

0 of 484

accepted fields wrong on 100 fresh forms (gate v2, fixed before the draw; Wilson 95% upper bound 0.79%)

A person re-read the AI-judged disputes blind: **0 read differently**.

AGAINST THE INCUMBENT · BAJA CALIFORNIA SUR, 2024 LOCAL ACTAS

91.6% field match, where the reading the state bought scored 62.84%.

Same per-field exact-match metric on Baja California Sur's 2024 local actas: ours on a pre-registered random 300, against the published values; theirs on all 2,020, against the manual capture.



Ours: 91.58% of 3,622 fields, Wilson 95% 90.63-92.44%. Theirs: 14,501 of 23,077 fields.

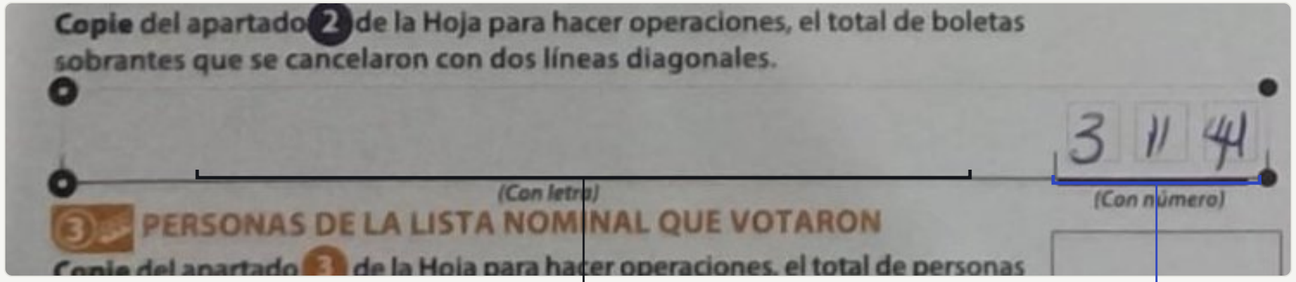
Same 300 forms, gate v1: Gemma accepts 1.8× the forms Qwen does, 4.6× cheaper, 2.8× faster.

READER, ON AMD	FORMS ACCEPTED	WRONG OF ACCEPTED FIELDS	$ PER 1,000 FORMS	TIME PER READ
Gemma-4-31B-IT	**55 of 300**	**2 of 1,210**	**$0.14**	**87 s**
Qwen3.8-27B	31 of 300	1 of 682	$0.64	241 s
Both must agree, the paper's sums as referee	89 of 300	**0 of 1,958**	$0.78	–

Rows 1-2: the first gate (v1) on the pre-registered 300. Row 3: two readers, a lighter gate, a secondary analysis. Gate v2 on 100 fresh forms: slide 7. Source: app/data/bakeoff_300.json.
Time per read: wall clock in the 300 run (2 in flight, our own pacing). Throughput, measured separately on the shared endpoint: 4.8 forms/min with 4 in flight, 32 s median (results/throughput_sweep.json).

$0.14 **Colima, a whole state:** all 982 located presidential sheets, read by Gemma on AMD.

Read 3114, paper 314. Gate v2, fixed in advance, sends it to a person.



Words line: blank
no second reading to compare

The paper: 314
in digits

Form 070212B0100, unused ballots (masked crop).

THE BLIND SPOT

The model read **3114**. Unused ballots sit in no identity: no sum can catch a digit inserted here. The first gate would have accepted it.

HELD OUT

Gate v2 (digits and words both read and equal) was fixed before 100 fresh forms were drawn. On them: **0 wrong of 484** accepted fields (Wilson 95% upper bound 0.79%).

SCOPE

By design, an audit of the typed values: most forms go to a person, and the auditor sees only the disputed fields. Not an official INE product.

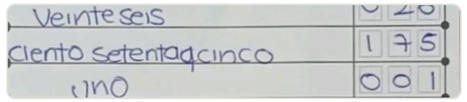
BUSINESS AT A GLANCE

An independent check of the keyed values. $1 per sheet audited (proposed).

Audit report: 6,358 fields checked, 267 disputed, each with its photo

PREP 2024, 300 pre-registered forms · 300 forms in the keyed file, 289 compared · audit `ab2b0555076b1` · 2026-10-17 · Merkle root `e56a8ef4f011b9e5` · fingerprint `c4788f2501814fd1`

3



In English: words: "ciento setenta y cinco" = 175 · boxes: 1 7 5

`050815B0100` · **Option 07** STRONG

`receipt`

`Judge this verdict yourself`

Keyed	Model digits	Model words
75	175	175

- model digits and words agree (175)
- the keyed value breaks rows sum to TOTAL (258 vs 358); the model's value satisfies it

The auditor looks at **267 of 6,358 fields (4%)**, each with its photo.

The product: the audit report, strongest first, sealed by one Merkle root.

BUYER

INE and the 32 state institutes, each with an outside auditor.

WHY NOW

6 June 2027: the federal midterm and 32 concurrent local elections. First step: a **paid shadow-mode pilot** on one state's count.

ONE STATE, PRICED

Morelos paid MXN 26.6M to run its 2024 PREP. Our check of every sheet: about **MXN 143k**, 0.5% of that. The federal count, 169,005 sheets: about $169,000.

NEXT

At least 7 more countries publish the same tally data; health reporting on DHIS2 in 70+ countries.

CHECK IT YOURSELF

Every number re-derives from public data.

REPRODUCE

```
make eval
```

Recomputes every number offline and checks it against what the site shows.

VERIFY A RECEIPT

```
python -m twohands.verify_receipt
  receipt.json --crop crop.jpg
```

A changed value or crop fails.

CHECK THE TIMESTAMP

```
ots verify
results/sample_audit_root.txt.ots
```

The report's Merkle root, anchored in Bitcoin with OpenTimestamps.

Two Hands. Verifiable quality control for handwritten numbers. `github.com/py-tr/twohands`